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AI-Powered University Club Management System

Project Roadmap & Development Plan

Full-Scale Web Platform Edition

Student Information

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Courses	Database Management Systems (DBMS) Software Engineering (SWE)
Type	Full-Scale Full Stack Web Application
Duration	11 Weeks (2025–2026)

Technology Stack

Frontend	HTML5, CSS3, Bootstrap 5, JS	Backend	Django 4.x (Python)
Database	MySQL 8.0	Auth	Django Allauth
Payments	SSLCommerz / bKash	AI	LLM API (Claude/OpenAI)
Deploy	Render / Railway	VCS	Git & GitHub

User Roles

ADMIN	Super Admin	PRESIDENT	Club President	MEMBER	Club Member	GUEST	Public User
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1. Project Overview

The **AI-Powered University Club Management System** is a full-scale, publicly accessible web platform for managing university club operations. Unlike a simple internal tool, this system has a **public-facing website** where guests can browse events and club information, and a **secure member portal** for registered users with role-based access.

Platform Architecture **Two distinct layers:**

- **Public Layer** — Landing page, event listings, blog, contact, online event registration & payment
- **Private Layer** — Member portal, admin dashboard, attendance, certificates, forum, analytics

What makes this project stand out: Real payment integration (SSLCommerz/bKash), public blog & forum, AI content generation, automated certificate delivery, and a fully deployed live URL.

2. Project Objectives

1. **Build a public website** with landing page, about, blog, and contact sections
2. **Enable online event registration** for both members and guests
3. **Integrate payment gateway** for event fees and membership dues
4. **Manage club members** with advanced role-based access control
5. **Automate attendance** tracking with QR code support
6. **Generate and email certificates** automatically post-event
7. **Provide a blog & news** platform managed by admin/president
8. **Host a discussion forum** for club members
9. **Deliver real-time analytics** through an advanced dashboard
10. **Integrate AI features** for event descriptions and blog drafts

3. Technology Stack

Layer	Technology	Purpose
Frontend	HTML5, CSS3	Structure and styling
UI Framework	Bootstrap 5	Responsive layouts, components
Interactivity	JavaScript ES6+	Dynamic UI, AJAX, form validation
Charts	Chart.js	Analytics dashboard graphs
Rich Text	TinyMCE / Quill	Blog and forum post editor
Backend	Django 4.x	Core application framework
Auth	Django Allauth	Social login, email verification
REST API	Django REST Framework	API endpoints for AJAX calls
Database	MySQL 8.0	Primary relational database
ORM	Django ORM	Database abstraction layer
Payment	SSLCommerz / bKash	Online payment processing
Email	Django Email + SMTP	Notifications, certificates
File Storage	Cloudinary / S3	Profile pictures, documents
PDF Generation	ReportLab / WeasyPrint	Certificate generation
QR Code	qrcode (Python)	Attendance QR codes
AI	LLM API	Content generation
Deployment	Render / Railway	Cloud hosting
Version Control	Git & GitHub	Source code management

4. User Roles & Permissions

Role	Access Level	Key Permissions
ADMIN Super Admin	Full system access	Manage all users, roles, payments, site settings
PRESIDENT Club President	Club-level management	Create events, approve members, post blog, manage forum
MEMBER Club Member	Personal portal	Register for events, pay fees, view certificates, join forum
GUEST Public Guest	Public pages only	Browse events, read blog, register & pay for public events

5. System Modules

5.1 Module 1 — Public Website

Public Layer
• Landing Page — Hero section, club highlights, stats counter, CTA buttons • About Page — Club history, mission, vision, team/executive members • Events Page — Publicly listed upcoming/past events with filter & search • Blog / News — Published articles, pagination, categories, tags • Contact Page — Contact form with email integration, Google Maps embed • Guest Event Registration — Public users can register & pay for events

5.2 Module 2 — Authentication & User Management

Auth System
• Registration with email verification • Login / Logout / Password Reset • Social Login (Google OAuth via Allauth) • Profile management (avatar, bio, contact) • Role assignment (Admin → President → Member) • Account activation / deactivation by Admin

5.3 Module 3 — Member Management

Member Portal

- Member directory with search, filter by department/batch/role
- Membership application and approval workflow
- Membership renewal with payment
- Member profile with activity history
- Export member list (CSV / Excel)

5.4 Module 4 — Event Management

Events System

- Create, Update, Delete events with rich details (CRUD)
- Event categories (Workshop, Seminar, Cultural, Sports, etc.)
- Public & member-only event visibility settings
- Online registration with seat limit management
- Event status: Upcoming / Ongoing / Completed / Cancelled
- Event gallery (image upload)
- AI-powered event description generation

5.5 Module 5 — Payment & Fee Management

Payment Gateway

- SSLCommerz / bKash payment integration
- Event registration fees (paid online)
- Annual membership fee collection
- Payment history and receipts (downloadable PDF)
- Admin payment dashboard with status (Paid / Pending / Failed)
- Refund request management

5.6 Module 6 — Attendance Management

Attendance System

- QR code-based attendance marking per event
- Manual attendance entry by President/Admin
- Attendance reports per event and per member
- Attendance percentage analytics
- Export attendance sheets (CSV / PDF)

5.7 Module 7 — Certificate Generation

Certificates

- Auto-generate PDF certificates on event completion
- Custom certificate templates per event type
- Unique certificate verification ID (QR code on cert)
- Member can download from personal portal
- Bulk certificate generation for all attendees
- Email certificate delivery option

5.8 Module 8 — Blog & News

Blog System

- Admin/President can create and publish articles
- Rich text editor (TinyMCE / Quill)
- Categories and tags
- Comment section for members
- Draft / Published / Archived status
- AI-assisted blog draft generation

5.9 Module 9 — Discussion Forum

Forum

- Members-only discussion boards
- Create threads, reply, like/upvote
- Categories: General, Announcements, Q&A, Projects
- Moderation tools (Admin/President can pin, delete)
- Search within forum

5.10 Module 10 — Dashboard & Analytics

Analytics Dashboard

- **Admin Dashboard:** Total members, revenue, events, attendance rate
- **President Dashboard:** Club-specific stats and activity feed
- **Member Dashboard:** Personal stats, upcoming events, certificates
- Charts: Member growth, event frequency, payment trends (Chart.js)
- Recent activity feed and notifications panel

5.11 Module 11 — AI Features

AI Integration

- **AI Event Description Generator** — Input title + type → Output polished description
- **AI Blog Draft Assistant** — Generate blog post drafts from a topic
- **Smart Email Notification** — AI-enhanced announcement messages
- All AI outputs are editable before saving

5.12 Module 12 — Notifications System

Notifications

- In-app notification bell (real-time updates)
- Email notifications for event registration, payment confirmation
- Automated reminders (event upcoming in 24h)
- Bulk announcement emails to all members

6. Database Design Overview

The database is designed in **Third Normal Form (3NF)** with the following core tables:

Table	Key Attributes
User	id, username, email, password_hash, role, is_active, created_at
Member	id, user_id (FK), full_name, dept, batch, roll, phone, avatar
Event	id, title, description, category, date, venue, capacity, fee, status, is_public
Registration	id, user_id (FK), event_id (FK), reg_type, registered_at
Attendance	id, member_id (FK), event_id (FK), status, marked_at
Payment	id, user_id (FK), event_id (FK), amount, method, txn_id, status
Certificate	id, member_id (FK), event_id (FK), cert_id (unique), file_path, issued_at
BlogPost	id, author_id (FK), title, content, category, status, published_at
Comment	id, post_id (FK), user_id (FK), body, created_at
ForumThread	id, author_id (FK), title, category, is_pinned, created_at
ForumReply	id, thread_id (FK), author_id (FK), body, upvotes, created_at
Notification	id, user_id (FK), message, type, is_read, created_at

7. Weekly Development Roadmap (11 Weeks)

Week 1 — Project Planning & Repository Setup

Goal: Establish project foundation and development environment.

Deliverables:

- Project Proposal document (PDF)
- GitHub repository with proper folder structure
- README.md with project overview and setup instructions
- Development environment setup guide (docs/setup.md)
- This Roadmap document (docs/roadmap.pdf)
- Initial .gitignore, requirements.txt

Tools: Git, GitHub, VS Code, Python 3.11+

Week 2 — Requirement Analysis & System Design

Goal: Document all system requirements and design high-level architecture.

Deliverables:

- Software Requirement Specification (SRS) document
- Functional Requirements (12 modules documented)
- Non-Functional Requirements (security, performance, scalability)
- Use Case Diagram (all 4 roles)
- System Architecture Diagram (3-tier: Client, Server, DB)
- Wireframes / UI Mockups (Figma or hand-drawn)

Week 3 — Database Design

Goal: Design a normalized, scalable database schema.

Deliverables:

- Entity-Relationship (ER) Diagram (12+ entities)
- Relational Schema with all FK relationships
- Database Normalization proof (1NF → 2NF → 3NF)
- Data Dictionary (field types, constraints, descriptions)

Week 4 — Database Implementation & Django Setup

Goal: Implement the database and configure the Django project.

Deliverables:

- MySQL database creation with all tables (schema.sql)
- Sample / seed data (seed.sql)
- Django project initialization with app structure

- Django Models mapped to all DB tables
- Migrations applied and verified
- Django Admin configured for all models

Django Apps: accounts, members, events, payments, attendance, certificates, blog, forum, notifications, core

Week 5 — Authentication & User Management

Goal: Build a robust multi-role authentication system.

Deliverables:

- User registration with email verification
- Login / Logout / Password Reset
- Google OAuth (via Django Allauth)
- Role-based access control (Admin, President, Member, Guest)
- User profile page (avatar upload, bio, contact update)
- Middleware for role-based page protection
- Admin panel for user role management

Week 6 — Public Website & Member Management

Goal: Build the public-facing website and member management features.

Deliverables:

- Landing Page (hero, highlights, stats, CTA)
- About Page (club history, executive team grid)
- Contact Page (form + email sending)
- Member directory with search and filter
- Membership application and approval workflow
- Member profile pages (public and private view)
- Responsive Bootstrap 5 design across all pages

Week 7 — Event Management & Online Registration

Goal: Full event lifecycle management with public registration.

Deliverables:

- Event CRUD (Create, Read, Update, Delete)
- Public events listing page (filter by category, date, status)
- Event detail page with registration form
- Seat limit management (auto-close on full)
- Member-only vs public event access control
- Event image gallery upload
- Event management dashboard (President/Admin)

Week 8 — Payment Gateway & Attendance

Goal: Integrate online payments and QR-based attendance.

Deliverables:

- SSLCommerz / bKash payment integration
- Event fee payment flow (registration → payment → confirmation)
- Membership fee payment
- Payment history and downloadable receipts (PDF)
- Admin payment dashboard (Paid / Pending / Failed)
- QR code generation per registered member per event
- QR code scanner for attendance marking
- Attendance reports (per event, per member, exportable)

Week 9 — Blog, Forum & Notifications

Goal: Build the community features and notification system.

Deliverables:

- Blog: create, publish, archive posts (rich text editor)
- Blog listing page with categories, tags, pagination
- Blog comments by members
- Forum: create threads, reply, upvote
- Forum moderation tools (pin, delete, lock thread)
- In-app notification system (bell icon, unread count)
- Email notifications (event reminders, payment confirmations)
- Bulk announcement email to all members

Week 10 — Certificate Generation, Dashboard & AI Features

Goal: Automate certificates, build analytics, and integrate AI.

Deliverables:

- Automated PDF certificate generation (ReportLab / WeasyPrint)
- Certificate template customization per event type
- Unique QR-based certificate verification system
- Bulk certificate generation + email delivery
- Admin Dashboard: revenue chart, member growth, event stats
- President Dashboard: club activity summary
- Member Dashboard: personal stats, event history, certs
- AI Event Description Generator (LLM API integration)
- AI Blog Draft Assistant
- Unit Testing (Django `TestCase`) for all modules
- System Testing (end-to-end user journeys)

Week 11 — Deployment, Security Hardening & Final Documentation

Goal: Launch the live system with full documentation.

Deliverables:

- Security audit (CSRF, XSS, SQL injection prevention)
- `DEBUG=False`, secret key in environment variables
- Static files on Cloudinary / WhiteNoise
- Live deployment on Render / Railway with MySQL
- Custom domain setup (optional)
- Final Project Report (PDF, 30+ pages)
- API Documentation (`docs/api.md`)
- Presentation Slides (PPT)
- Project demonstration video (screen record)
- Final GitHub cleanup and tag (`v1.0.0`)

8. Project Timeline Summary

Wk	Phase	Primary Modules	Key Output
1	Planning	Setup	Repo, Roadmap
2	Analysis	Requirements	SRS, Wireframes, Architecture
3	Design	Database	ER Diagram, Schema
4	Foundation	DB + Django	Models, Migrations, Admin
5	Backend	Auth	Login, Roles, Profiles
6	Backend	Public + Members	Landing Page, Member Directory
7	Backend	Events	Event CRUD, Public Registration
8	Backend	Payments + Attendance	Payment Gateway, QR Attendance
9	Backend	Blog + Forum + Notifications	Community Features
10	Full Stack	Certs + Dashboard + AI	Analytics, AI, Testing
11	Launch	Deployment	Live URL, Final Docs

9. Repository Structure

Recommended GitHub Folder Structure

```
club-management-system/
|-- docs/
|   |-- roadmap.pdf
|   |-- roadmap.tex
|   |-- SRS.pdf
|   |-- ER_Diagram.png
|   |-- architecture.png
|   +-- api.md
|-- backend/
|   |-- config/           (settings, urls, wsgi)
|   |-- accounts/        (auth, profiles)
|   |-- members/         (member management)
|   |-- events/          (event CRUD, registration)
|   |-- payments/        (payment gateway)
|   |-- attendance/      (QR, marking, reports)
|   |-- certificates/    (PDF generation)
|   |-- blog/            (posts, comments)
|   |-- forum/           (threads, replies)
|   |-- notifications/   (email, in-app)
|   |-- ai_features/     (LLM API integration)
|   +-- core/            (landing page, about, contact)
|-- frontend/
|   |-- static/          (css, js, images)
|   +-- templates/       (HTML templates)
|-- database/
|   |-- schema.sql
|   +-- seed.sql
|-- requirements.txt
|-- .env.example
|-- .gitignore
+-- README.md
```

10. Expected Outcomes

- A live, publicly accessible full-scale web application
- Real payment processing with transaction history
- QR-based attendance and automated certificate delivery
- Community features (blog, forum) with moderation
- Advanced 4-role access control system

- AI-powered **content generation** integrated into the workflow
- **Professional-grade codebase** on GitHub with documentation
- Demonstrates mastery of **DBMS, SWE, Full-Stack, and AI integration**

Deployment Target: The final system will have a live URL that can be shared with faculty, peers, and included in a portfolio or CV.

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